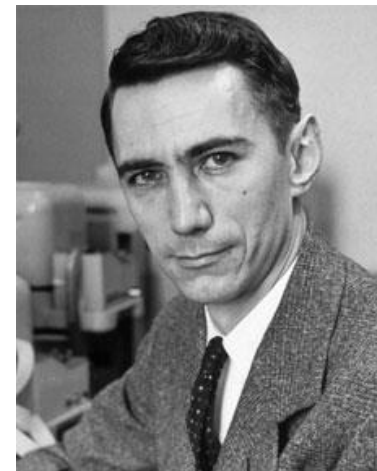


Lecture 16

Entropy, Information and Logistic Regression

Reference Book Chapters: Chapter 2.5



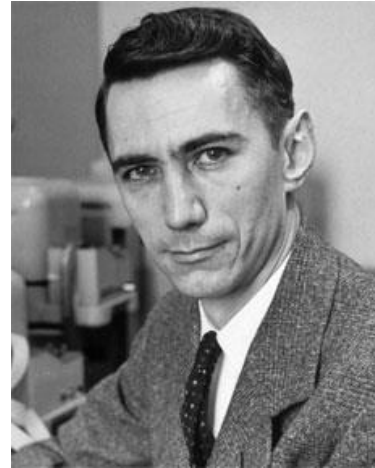
Entropy: How to measure information

- Rain in Seattle: Say each day we flip a coin and it rains with 80% probability, independently. (A rain random variable X).
- We observe a sequence of 100 days:
110101011011101111010111...101111
- Q: How much information is there in this sequence?
- Not 100 bits ?
- How much can we compress it?
- Shannon's compression theorem: Entropy is the ultimate limit to compression

Setup for Shannon's compression theorem



Alan



Claude

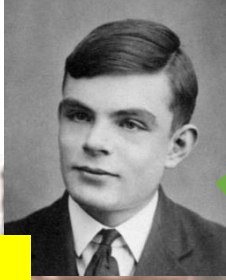
First they meet before and **they know the statistics of the rain random variable X , but not the realizations.**

They agree on some compression scheme:

When I see 1111, I will send 01, otherwise I will send 10 followed by the weather in the next 5 days. ...

Setup for Shannon's compression theorem

1101110110...



Alan

011010



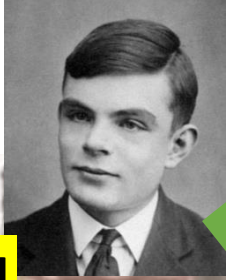
Claude

1101110110...



Always Rain = 0 information=compressible to 0 bits

111111111111



Alan

duh



Claude

1111111111



Definition of entropy

Entropy definition

$$H(X) = - \sum_x p(x) \log p(x)$$

Entropy of a RV:
of bits needed to compress this RV
with distribution p

Bernalli R.V. : $X_B \begin{cases} 1 & \text{with probability } p. \\ 0 & \text{w.p. } 1-p. \end{cases}$

- $H(\text{Rain}) = - [p \log (p) + (1-p) \log (1-p)]$

■ Use log base 2. Also set $0 \log 0 = 0$, by continuity.

Entropy is the limit to compression

- $H(\text{Rain}) = - [p \log (p) + (1-p) \log (1-p)]$
- If $p=0$, $H(\text{Never Rain}) = - [0 \log 0 + 1 \cdot \log 1] = 0$ bits. (perfectly compressible)
- if $p=1$, $H(\text{Always Rain}) = - [1 \log 1 + 0 \log 0] = 0$ bits (perfectly compressible)
- if $p=0.8$ $H(\text{Rain}) = - [0.8 \log 0.8 + 0.2 \cdot \log 0.2] = 0.7219...$ bits per day
- Shannon's compression theorem:
we can compress 100 bits of rain data into **73 bits**
- if $p=0.9$, $H(\text{Rain}) = 0.468...$, so we can further compress 100 bits of rain data into **47 bits**. Less surprise = Less information = more compression.
- Only when $p=1/2$ (fair coin) we have maximum surprise = maximum information = no compression is possible.
- $H(\text{fair coin}) = 1$ bit per realization.

Intuitive understanding of the compression theorem

- Fact 1: To describe one object in a list of L things, I need $\log L$ bits

Intuitive understanding of the compression theorem

- Fact 1: To describe one object in a list of L things, I need $\log L$ bits

$L = 4$

Banana

Orange

Apple

Mango

Intuitive understanding of the compression theorem

- Fact 1: To describe one object in a list of L things, I need $\log L$ bits

$L = 4$

Banana 00

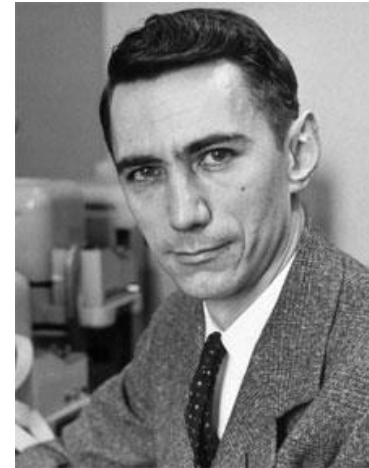
Orange 01

Apple 10

Mango 11



Alan



Claude

Intuitive understanding of the compression theorem

- Fact 1: To describe one object in a list of L things, I need $\log L$ bits
- Fact 2: When compressing random sequences, there are only a few sequences you need to represent. Only those will appear with very high probability. We call them typical sequences.
- We only need $\log (\text{number of typical sequences})$ bits to compress and succeed with high probability.

Intuitive understanding of the compression theorem

- When I flip a coin 100 times with $p=0.10$ I expect to see about 10 heads.
- Alan and Claude will only worry about representing those sequences with exactly 10 heads. If it rains fewer or more than 10 times in 100 days they will declare failure.
- How many ways are there to place 10 sunny days in 100 positions?

Intuitive understanding of the compression theorem

- When I flip a coin 100 times with $p=0.10$ I expect to see about 10 heads.
- Alan and Claude will only worry about representing those sequences with exactly 10 heads. If it rains fewer or more than 10 times in 100 days they will declare failure.
- How many ways are there to place 10 sunny days in 100 positions?
- How about placing 90 sunny days in 100 positions?

Intuitive understanding of the compression theorem

- When I flip a coin 100 times with $p=0.10$ I expect to see about 10 heads.
- Alan and Claude will only worry about representing those sequences with exactly 10 heads. If it rains fewer or more than 10 times in 100 days they will declare failure.
- How many ways are there to place 10 sunny days in 100 positions?

$$L = \binom{100}{10} \longrightarrow \log L \approx 100H(p)$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} \approx 2^{nH(k/n)}$$

Intuitive understanding of the compression theorem

- Fact 1: To describe one object in a list of L things, I need $\log L$ bits
- Fact 2: When compressing random sequences, there are only a few sequences you need to represent. Only those will appear with very high probability. We call them typical sequences.
- We only need $\log(\text{number of typical sequences})$ bits to compress and succeed with high probability.
- Entropy of any random variable is essentially equal to the logarithm of the number of typical sequences. (Asymptotic Equipartition Property).
- This is essentially the Law of Large numbers.

Horse race problem

Horse name. Probability

1	$1/2$
2	$1/4$
3	$1/8$
4	$1/16$
5	$1/64$
6	$1/64$
7	$1/64$
8	$1/64$

One of the 8 horses wins. How many bits do you need to send to communicate the outcome to the race?

(3 bits is naïve). You can do better.

Horse race problem

Horse name. Probability Naïve code

1	$1/2$	000	0
2	$1/4$	001	10
3	$1/8$	010	110
4	$1/16$	011	1110
5	$1/64$	100	111101
6	$1/64$	101	111110
7	$1/64$	110	111111
8	$1/64$	111	111100

One of the 8 horses wins. How many bits do you need to send to communicate the outcome to the race?

(3 bits is naïve). You can do better.

You can compute the entropy of this random variable and see if this compression code is optimal or not (i.e. matching the Shannon Limit).

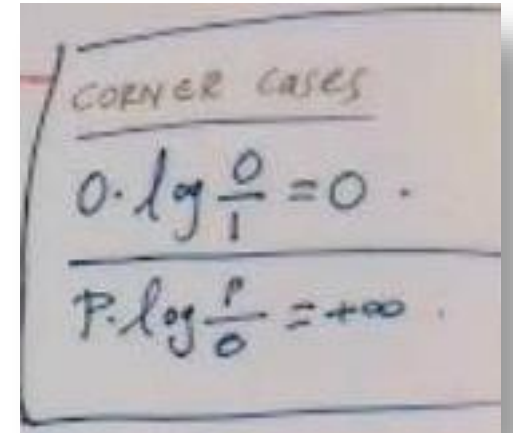
More details: Entropy and Information theory

- How to compute entropy from distributions and datasets.
- What is KL Divergence and Cross Entropy
- Link between Cross Entropy, KL Divergence and Entropy

Entropy

Entropy definition

$$H(X) = - \sum_x p(x) \log p(x)$$

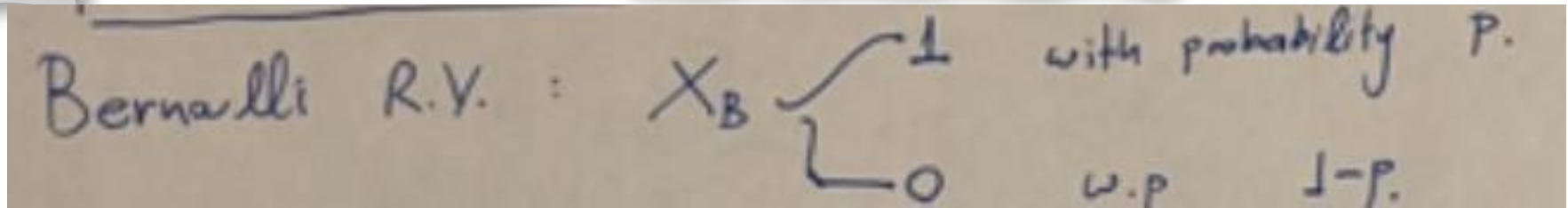


CORNER CASES

$$0 \cdot \log \frac{0}{1} = 0$$
$$P \cdot \log \frac{P}{0} = +\infty$$

Example:

Random Variable 1:
Fair Coin



Bernoulli R.V. : $X_B \begin{cases} 1 & \text{with probability } P. \\ 0 & \text{w.p. } 1-P. \end{cases}$

KL Divergence and Cross Entropy

Kullback-Leibler
Divergence
 $D(p \parallel q)$

$$D(p \parallel q) = \sum_x p(x) \cdot \log \frac{p(x)}{q(x)}$$

CORNER CASES

$$0 \cdot \log \frac{0}{1} = 0$$
$$P \cdot \log \frac{P}{0} = +\infty$$

Example:

$p = \begin{cases} 1 & \text{w.p. } 1/2 \\ 0 & \text{w.p. } 1/2 \end{cases}$
$q = \begin{cases} 1 & \text{w.p. } 1 \\ 0 & \text{w.p. } 0 \end{cases}$

Random Variable 1:
Fair Coin

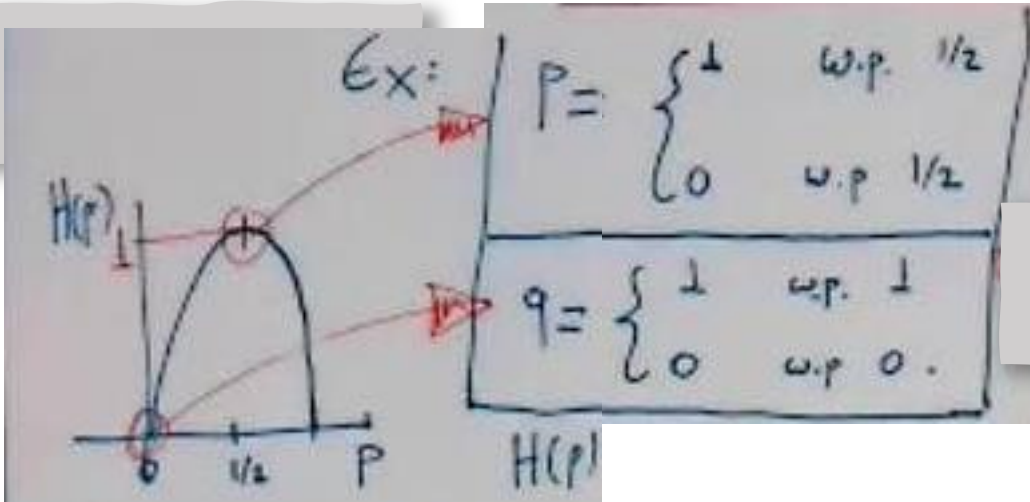
Random Variable 2:
Deterministic

$H(p) = ?$
 $H(q) = ?$

Q: Compute KL Divergence
between p and q

KL Divergence and Cross Entropy

Kullback-Leibler
Divergence



Random Variable 1:
Fair Coin

Random Variable 2:
Deterministic

Q: Compute KL Divergence
between p and q

CORNER CASES

$$0 \cdot \log \frac{0}{1} = 0$$
$$P \cdot \log \frac{P}{0} = +\infty$$

$H(p) = ?$
 $H(q) = ?$

KL Divergence and Cross Entropy

Kullback-Leibler
Divergence

$$D(p||q) = \sum_x p(x) \cdot \log \frac{p(x)}{q(x)}$$

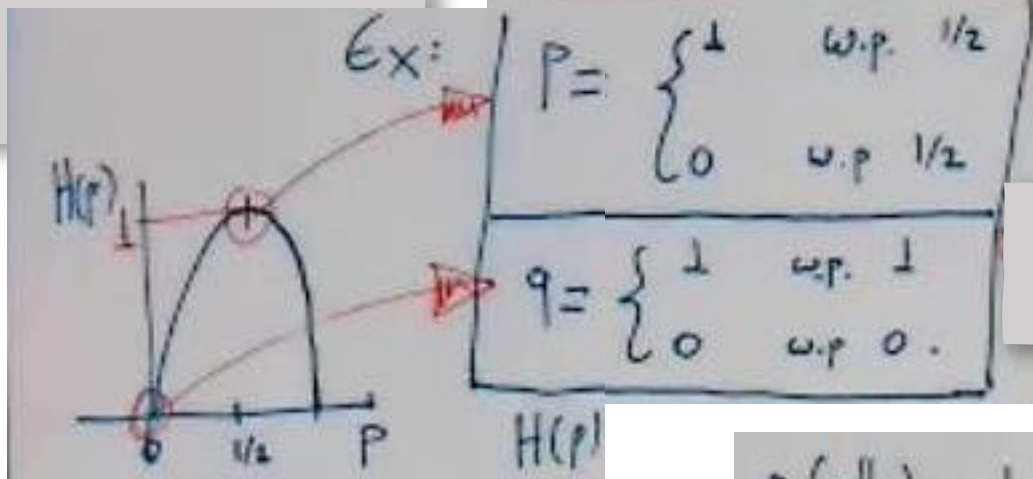
CORNER CASES

$$0 \cdot \log \frac{0}{1} = 0$$
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Random Variable 1:
Fair Coin

Random Variable 2:
Deterministic

$H(p)=?$
 $H(q)=?$



Q: Compute KL Divergence
between p and q

$$D(p||q) = \frac{1}{2} \cdot \underbrace{\log \frac{1/2}{0}}_{+\infty} + \frac{1}{2} \cdot \underbrace{\log \frac{1/2}{1}}_{-1} = +\infty$$

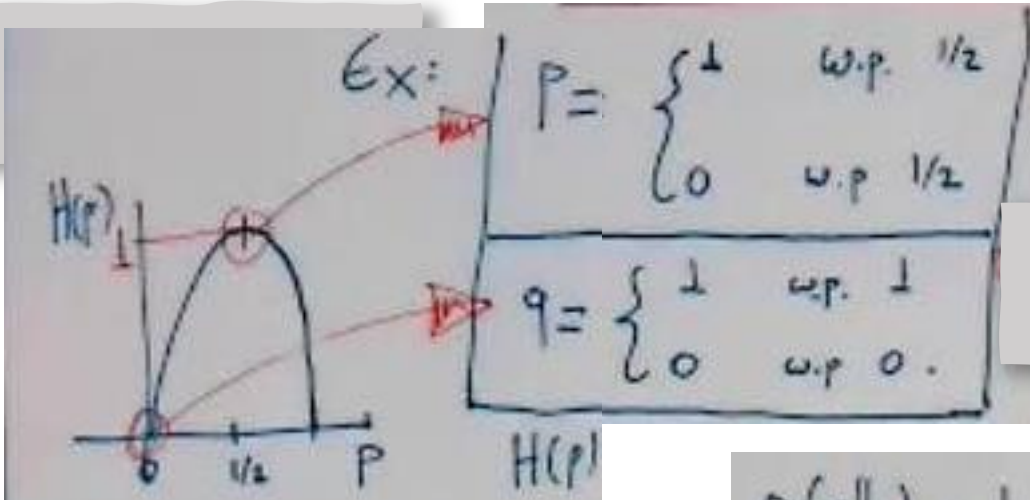
KL Divergence and Cross Entropy

Kullback-Leibler
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$$D(p||q) = \sum_x p(x) \cdot \log \frac{p(x)}{q(x)}$$

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Random Variable 1:
Fair Coin

Random Variable 2:
Deterministic

$H(p)=?$
 $H(q)=?$

Q: Compute KL Divergence
between p and q

$$D(p||q) = \frac{1}{2} \cdot \underbrace{\log \frac{1/2}{0}}_{+\infty} + \frac{1}{2} \cdot \underbrace{\log \frac{1/2}{1}}_{-1} = +\infty$$
$$D(q||p) = \underbrace{1 \cdot \log \frac{1}{1/2}}_1 + \underbrace{0 \cdot \log \frac{0}{1/2}}_0 = 1$$

KL Divergence and Cross Entropy

Cross Entropy

For two distributions (PMFs) p, q :

$$H(p, q) = - \sum_x p(x) \cdot \log q(x).$$

Cross Entropy definition

Cross Entropy

For two distributions (PMFs) p, q :

$$H(p, q) = - \sum_x p(x) \cdot \log q(x).$$

ex: $H(p, q)$
 $H(q, p) =$

$p = \begin{cases} 1 & \text{w.p. } 1/2 \\ 0 & \text{w.p. } 1/2 \end{cases}$	Fair coin
$q = \begin{cases} 1 & \text{w.p. } 1 \\ 0 & \text{w.p. } 0 \end{cases}$	Deterministic quantity

Cross Entropy

Cross Entropy

For two distributions (PMFs) p, q :

$$H(p, q) = - \sum_x p(x) \cdot \log q(x).$$

ex: $H(p, q) = - \left[\frac{1}{2} \cdot \log 1 + \frac{1}{2} \cdot \log 0 \right] = +\infty.$

$H(q, p) = - \left[1 \cdot \log \frac{1}{2} + 0 \cdot \log \frac{1}{2} \right] = 1.$

$p = \begin{cases} 1 & \text{w.p. } 1/2 \\ 0 & \text{w.p. } 1/2 \end{cases}$	Fair coin
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Sanity check

$H(p, p) = ?$

Cross Entropy

Cross Entropy

For two distributions (PMFs) p, q :

$$H(p, q) = - \sum_x p(x) \cdot \log q(x).$$

ex: $H(p, q) = - \left[\frac{1}{2} \cdot \log 1 + \frac{1}{2} \cdot \log 0 \right] = +\infty.$

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$p = \begin{cases} 1 & \text{w.p. } 1/2 \\ 0 & \text{w.p. } 1/2 \end{cases}$	Fair coin
$q = \begin{cases} 1 & \text{w.p. } 1 \\ 0 & \text{w.p. } 0. \end{cases}$	Deterministic quantity

Sanity check
 $H(p, p) = H(p)$

Let's connect KL Divergence and Cross Entropy

$$D(p||q) = \sum_x p(x) \cdot \log \frac{p(x)}{q(x)}$$

Recall

$$\log \frac{a}{b} = \log a - \log b.$$

$$\log a \cdot b = \log a + \log b.$$

Let's connect KL Divergence and Cross Entropy

$$D(p||q) = \sum_x p(x) \cdot \log \frac{p(x)}{q(x)}$$

$$= \underbrace{\sum_x p(x) \cdot \log p(x)}_{-H(p)} - \underbrace{\sum_x p(x) \cdot \log q(x)}_{H(p,q)}$$

$$D(p||q) = -H(p) + H(p,q) \Rightarrow H(p,q) = D(p||q) + H(p)$$

Total # of bits needed to describe a RV with distribution p , if your compression scheme was designed for q

Extra # of bits wasted because you used q for p

Total # of bits needed to compress a RV with distribution p

Recall

$$\log \frac{a}{b} = \log a - \log b$$

$$\log a \cdot b = \log a + \log b$$

Let's connect KL Divergence and Cross Entropy

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Recall

$$\log \frac{a}{b} = \log a - \log b$$

$$\log a \cdot b = \log a + \log b$$

Linking KL and Cross entropy example

$$P = \begin{cases} 1 & \text{up } 1/2 \\ 0 & \text{up } 1/2 \end{cases}$$

Fair coin.

$$q = \begin{cases} 1 & \text{up } 1 \\ 0 & \text{up } 0 \end{cases}$$

Deterministic.

Recall

$$D(p \parallel q) = \sum_x p(x) \cdot \log \frac{p(x)}{q(x)}.$$

$$H(p, q) = - \sum_x p(x) \cdot \log q(x).$$

$$\text{Link: } H(p, q) = D(p \parallel q) + H(p)$$

$$H(p, q) = ?$$

$$D(p \parallel q) = ?$$

$$H(p) = ?$$

Linking KL and Cross entropy example

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$$D(p \parallel q) = \sum_x p(x) \cdot \log \frac{p(x)}{q(x)}$$

$$H(p, q) = - \sum_x p(x) \cdot \log q(x)$$

Link: $H(p, q) = D(p \parallel q) + H(p)$

You cannot compress p
using a scheme for q !

$$H(p, q) = \infty$$

Total # of bits needed to
describe a RV with
distribution p , if your
compression scheme was
designed for q

$$D(p \parallel q) = \infty$$

Extra # of bits wasted
because you used q for p

$$H(p) = 1$$

Total # of bits needed to
compress a RV with
distribution p

CORNER CASES

$$0 \cdot \log \frac{0}{1} = 0$$

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Linking KL and Cross entropy example

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$$D(p \parallel q) = \sum_x p(x) \cdot \log \frac{p(x)}{q(x)}$$

$$H(p, q) = - \sum_x p(x) \cdot \log q(x)$$

Link: $H(p, q) = D(p \parallel q) + H(p)$

$$H(p, q) = \infty$$

Total # of bits needed to describe a RV with distribution p , if your compression scheme was designed for q

$$D(p \parallel q) = \infty$$

Extra # of bits wasted because you used q for p

$$H(p) = 1$$

Total # of bits needed to compress a RV with distribution p

Linking KL and Cross entropy example

$$P = \begin{cases} 1 & \text{up } 1/2 \\ 0 & \text{up } 1/2 \end{cases}$$

Fair coin.

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Deterministic.

Recall

$$D(p \parallel q) = \sum_x p(x) \cdot \log \frac{p(x)}{q(x)}$$

$$H(p, q) = - \sum_x p(x) \cdot \log q(x)$$

$$\text{Link: } H(q, p) = D(q \parallel p) + H(q)$$

$$H(q, p) = ?$$

Total # of bits needed to describe a RV with distribution p , if your compression scheme was designed for q

$$D(q \parallel p) = ?$$

Extra # of bits wasted because you used q for p

$$H(q) = ?$$

Total # of bits needed to compress a RV with distribution p

Linking KL and Cross entropy example

$$P = \begin{cases} 1 & \text{up } 1/2 \\ 0 & \text{up } 1/2 \end{cases}$$

Fair coin.

$$q = \begin{cases} 1 & \text{up } 1 \\ 0 & \text{up } 0 \end{cases}$$

Deterministic.

Recall

$$D(p \parallel q) = \sum_x p(x) \cdot \log \frac{p(x)}{q(x)}$$

$$H(p, q) = - \sum_x p(x) \cdot \log q(x)$$

$$\text{Link: } H(q, p) = D(q \parallel p) + H(q)$$

$$H(q, p) = 1$$

Total # of bits needed to describe a RV with distribution p , if your compression scheme was designed for q

$$D(q \parallel p) = 1$$

Extra # of bits wasted because you used q for p

$$H(q) = 0$$

Total # of bits needed to compress a RV with distribution p

Recap

- We learned the definitions of KL Divergence, Entropy and Cross Entropy.
- How to compute them for simple distributions
- Their connection $H(p,q) = D(p \parallel q) + H(p)$
- Next thing we will learn: Cross entropy will be the loss we use for logistic regression and deep learning.

Remembering Logistic Regression

- Logistic regression **is not a regression**.
- It is a method for **binary classification**, i.e. the target variable y is 0 or 1.
- To recall from L10:
Logistic Regression directly models $p(y|x)$ by the sigmoid. Does not assume data x is gaussian.
- If you also assume x is gaussian you get?
If you assume the data x is gaussian and has the same covariance you get?

Logistic Regression

- Logistic regression **is not a regression**.
- It is a method for **binary classification**, i.e. the target variable y is 0 or 1.
- To recall from L10:
Logistic Regression directly models $p(y|x)$ by the sigmoid. Does not assume data x is gaussian.
- If you also assume x is Gaussian you get? Gaussian Discriminant Analysis (QDA).
If the covariances are different, its also called QDA.
- If the covariances are the same, its called LDA.

Logistic Regression

- Logistic regression **is not a regression**.
- It is a method for **binary classification**, i.e. the target variable y is 0 or 1.
- To recall from L10:
Logistic Regression directly models $p(y|x)$ by the sigmoid. Does not assume data x is gaussian.
- If you also assume x is Gaussian you get? Gaussian Discriminat Analysis (QDA).
If the covariances are different, its also called QDA.
- If the covariances are the same, its called LDA.

Working example:

Predict if a customer will click on a Home insurance Ad given two features we know about them.

Logistic Regression

Predict if a customer will click on a Home insurance Ad given two features we know about them.

Customer Features

Age	If own CAR	if own Home	Y (did click)
30	0	0	0
45	1	0	1

Logistic Regression model:

Predict probability that a customer clicks on ad given their features

$x_1 = [30, 0, 0]$

Logistic Regression

Predict if a customer will click on a Home insurance Ad given two features we know about them.

Customer Features

Age	If own CAR	if own Home	Y (did click)
30	0	0	0
45	1	0	1

Sigmoid function:

$$S(z) = \frac{1}{1 + \exp(-z)}$$

Sigmoid converts score to a probability
(something from 0 to 1)

Logistic Regression model:
 $\text{Prob}(Y=1 \text{ for some } x) = S(z)$
 $= S(w^T x)$

Logistic Regression

Predict if a customer will click on a Home insurance Ad given two features we know about them.

Customer Features

Age	If own CAR	if own Home	Y (did click)
30	0	0	0
45	1	0	1

Say we have customer 1: $x=[30,0,0]$.
Compute their score and probability to click

Sigmoid function:

$$S(z) = \frac{1}{1 + \exp(-z)}$$

Logistic Regression model:
 $\text{Prob}(Y=1 \text{ for some } x) = S(z)$
 $= S(w^T x)$

Logistic Regression

Predict if a customer will click on a Home insurance Ad given two features we know about them.

Customer Features

Age	If own CAR	If own Home	Y (did click)
30	0	0	0
45	1	0	1

Say we have customer 1: $x = [30, 0, 0]$.

Compute a score $z = w_1 * x[1] + w_2 * x[2] + w_3 * x[3]$

$$z = w^T x$$

Sigmoid function:

$$S(z) = \frac{1}{1 + \exp(-z)}$$

Logistic Regression model:

$$\text{Prob}(Y=1 \text{ for some } x) = S(z)$$

$$= S(w^T x)$$

Logistic Regression

Predict if a customer will click on a Home insurance Ad given two features we know about them.

Example: For Customer 2. $x_2 = [45, 0, 1]$.

Compute the probability of this customer clicking on the Ad, for a logistic regression with parameters $w = [0.1, 1, 10]$

Solution: 1. Compute the score $z_2 = w^T x_2$.

$$z_2 = 0.1 * 45 + 1 * 0 + 10 * 1 = 14.5.$$

2. Use the logistic regression model: $\text{Prob}(Y=1 \text{ for } x_2) = S(w^T x_2)$
 $= 1 / (1 + \exp(-14.5)) = 0.9999$

So this logistic regression thinks that customer 2 will click with probability 0.9999.

The training data says that $y_2 = 1$ indeed.

Sigmoid function:

$$S(z) = \frac{1}{1 + \exp(-z)}$$

Logistic Regression model:
 $\text{Prob}(Y=1 \text{ for some } x) = \text{Sigmoid}(z)$
 $= \text{Sigmoid}(w^T x)$

The likelihood of the data

- The logistic regression model **dreams** of some probability for each customer clicking or not the Ad. But we know what actually happened in the training data.
- Likelihood of the data= the probability of the data happening under the logistic regression probabilities.
- **Example 3:** For this dataset, and model $w=[0.1,1,10]$ find the likelihood of the data.
- $L(w) = P(\text{Customer 1 not clicking}) * P(\text{Customer 2 clicking})$
- $\text{Prob}(\text{Customer 2 clicking}) = S(w * x_2) = 0.999$
- $\text{Prob}(\text{Customer 1 not clicking}) = 1 - \text{Prob}(\text{Customer 1 clicking}) = 1 - S(w * x_1)$
- $= 1 - \text{sigmoid}(30 * 0.1 + 0 * 1 + 0 * 10) = 1 - S(3) = 0.047$
- So Likelihood of the data for this w is $L(w) = 0.999 * 0.047 = 0.046$
- pretty low.

Customer Features

Age	If OWN CAR	if OWN Home	Y (did click)
30	0	0	0
45	1	0	1

The likelihood of the data

- Likelihood of the data= the probability of the data happening under the logistic regression probabilities. General calculation.
- For each sample i , if $y_i=1$ we want $S(z_i)$
- if $y_i= 0$, we want $1- S(z_i)$
- This can be compactly written as $(S(z_i))^{y_i} (1-S(z_i))^{1-y_i}$
- So for n training samples the likelihood is
- $L(w)=$ Product over all i : $(S(z_i))^{y_i} (1-S(z_i))^{1-y_i}$
- $\text{Log } L(w)=$ Sum over all i : $y_i \text{Log } S(z_i) + (1-y_i)* \text{Log } 1- S(z_i)$
- Lets call $\mu_i= S(z_i)$, Then $\text{Log } L(w)=$ Sum over all i : $y_i \text{Log } \mu_i + (1-y_i)* \text{Log } 1- \mu_i$
- So $\text{Log } L(w)$ is the **minus** cross entropy between two distributions: $y= [y_i, 1-y_i]$ and $\mu= [\mu_i, 1- \mu_i]$
- $\text{Log } L(w) = -$ Sum over all i : $H([y_i, 1-y_i], [\mu_i, 1- \mu_i])$
- Negative Log Likelihood = Cross Entropy $([y_i, 1-y_i], [\mu_i, 1- \mu_i])$.

Customer Features

Age	If own CAR	if own Home	Y (did click)
30	0	0	0
45	1	0	1

The likelihood of the data

- Likelihood of the data= the probability of the data happening under the logistic regression probabilities. General calculation.
- For each sample i , if $y_i=1$ we want $S(z_i)$
- if $y_i= 0$, we want $1- S(z_i)$
- This can be compactly written as $(S(z_i))^{y_i} (1-S(z_i))^{1-y_i}$
- So for n training samples the likelihood is
- $L(w)=$ Product over all i : $(S(z_i))^{y_i} (1-S(z_i))^{1-y_i}$
- $\text{Log } L(w)=$ Sum over all i : $y_i \text{ Log } S(z_i) + (1-y_i)* \text{Log } 1- S(z_i)$
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Customer Features

Age	If own CAR	if own Home	Y (did click)
30	0	0	0
45	1	0	1

Cross Entropy

$$H(p, q) = -[p \log(q) + (1 - p) \log(1 - q)]$$

Recap: Logistic Regression

- Logistic Regression is a model for binary classification
- Produces soft scores for classes 0,1 by performing a linear regression to compute a score and then that score is converted to a probability through the logistic function.
- To find the best logistic regression parameters we maximize the likelihood.
- Negative Log Likelihood = Cross Entropy ($[y_i, 1-y_i]$, $[\mu_i, 1- \mu_i]$).

How to do multi-class classification?

Multi-class logistic regression.

sigmoid-> Softmax

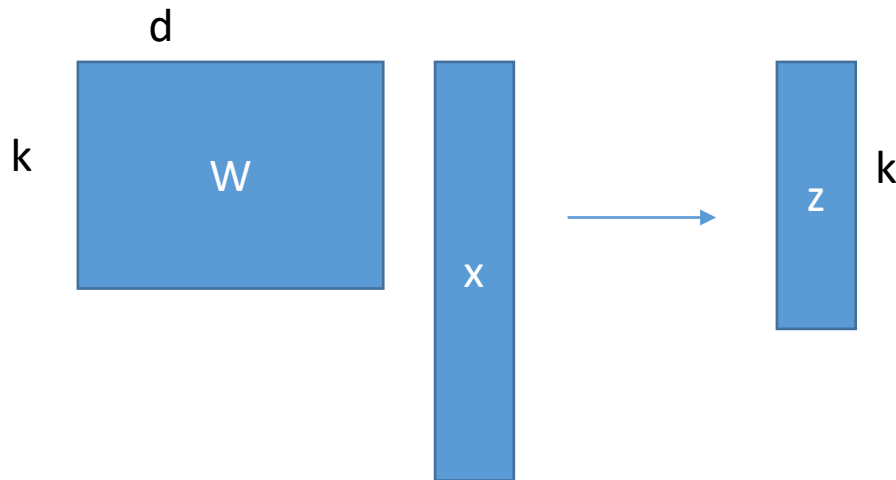
- Given a vector of scores $z = [1, 2, 3]$ the softmax is another vector s
- $s_i = \exp(z_i) / C$
- where $C = \sum \exp(z_i)$
- $\exp(1)=2.7$, $\exp(2)= 7.39$, $\exp(3)= 20.08$

So $s = [\exp(1) \exp(2) \exp(3)] / 30.19 = [0.09, 0.24, 0.66]$

Exercise, compute softmax of z where $z = [-1, 0, 1]$

multi-class logistic regression

- Given d features x , and k classes, you have a matrix of weights that is k by d .

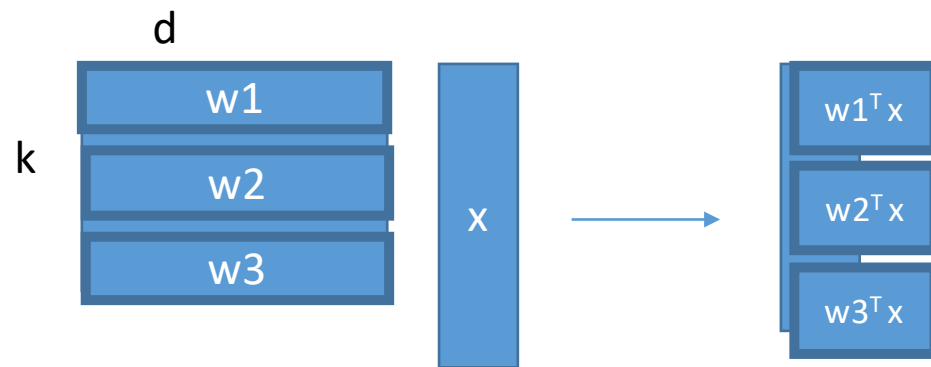


- Final probabilities are $\text{softmax}(z) = \text{softmax}(Wx)$

multi-class logistic regression

- Given d features x , and $k=3$ classes, you have a matrix of weights that is k by d .

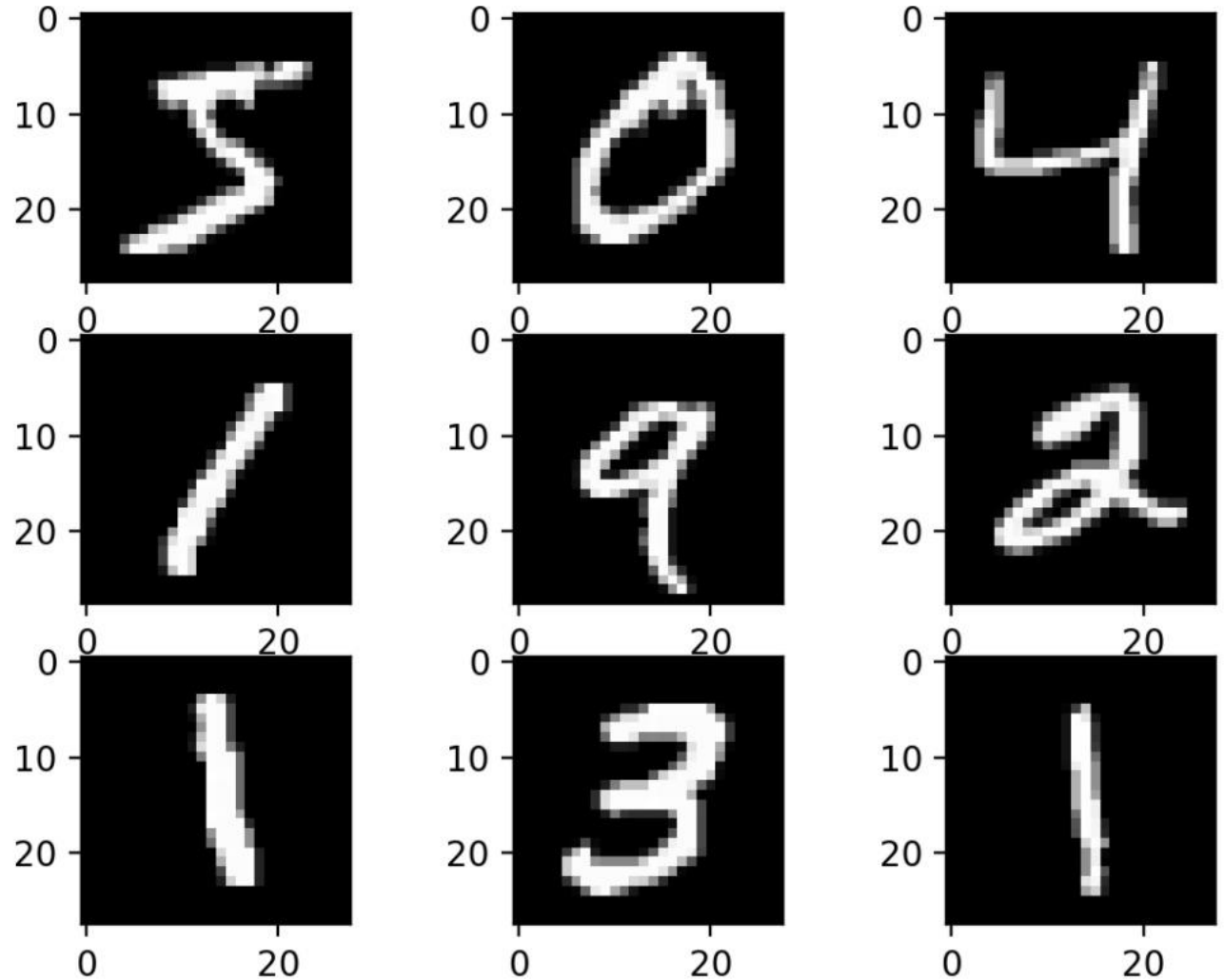
$k=3$ so we have 3 classes.



- Final probabilities are $\text{softmax}(z) = \text{softmax}(W x)$

MNIST dataset

- 60000 handwritten digits
- 28x28 pixels,
- black and white.



CIFAR10 dataset



CIFAR10 dataset

- 60k 32x32 colour images in 10 classes,
- 6k images per class.
- 50k training images and 10k test images.
- Divided into five training batches and one test batch (each with 10k images)
- The test batch: 1000 randomly-selected images from each class.
- The training batches contain the remaining images in random order,
- Some training batches may contain more images from one class than another. Between them, the training batches contain exactly 5000 images from each class.

airplane



automobile



bird



cat



deer



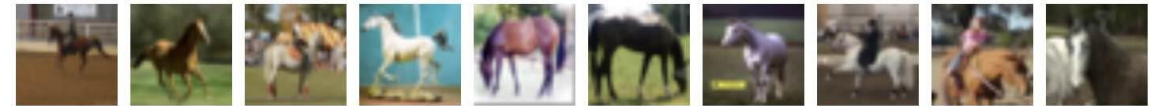
dog



frog



horse



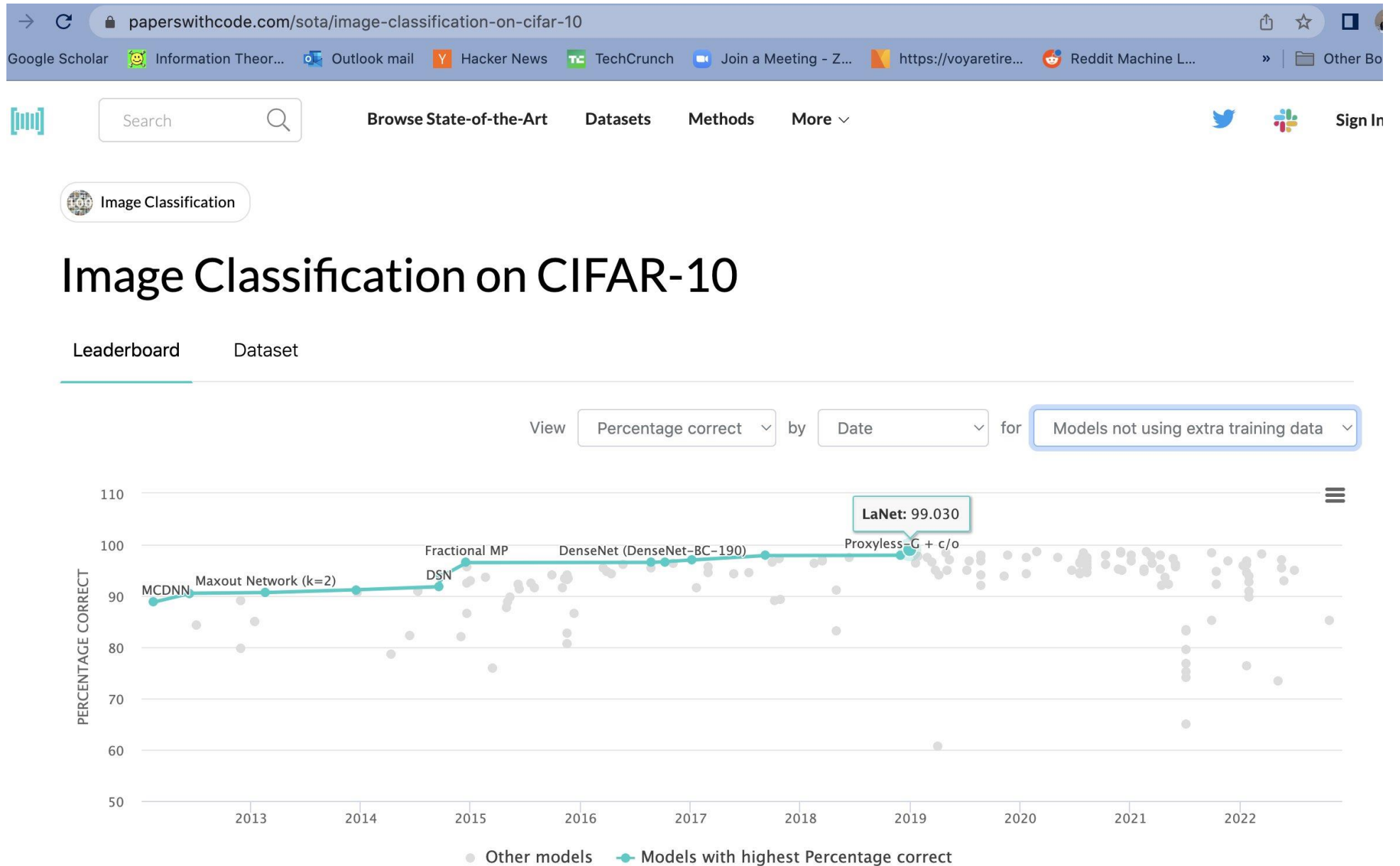
ship



truck

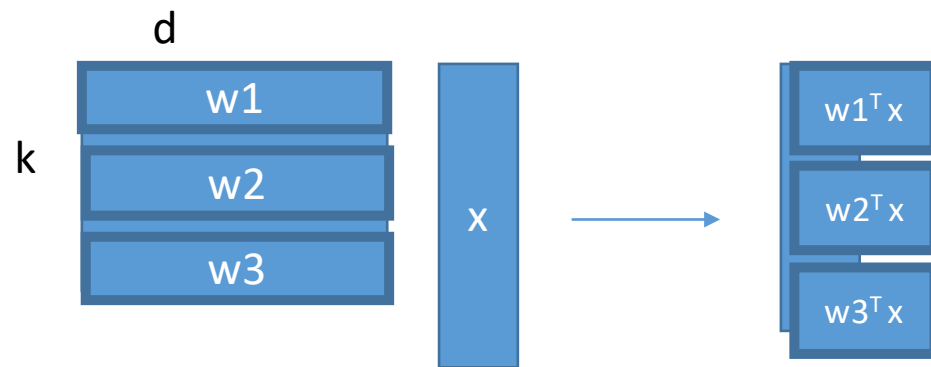


What is SOTA



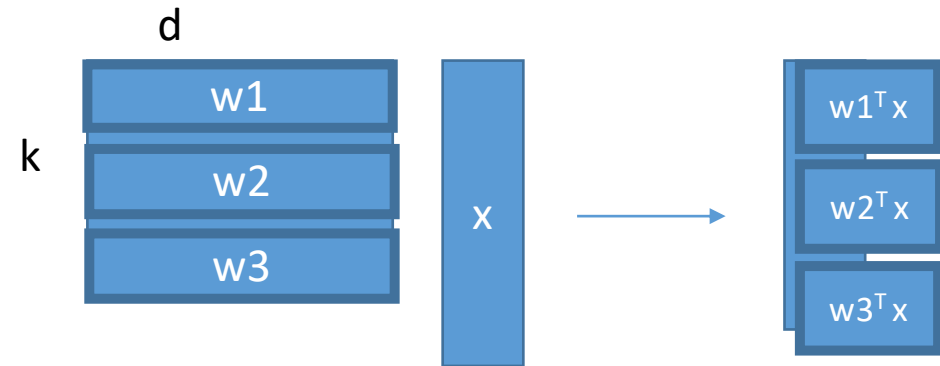
multi-class logistic regression

- Given d features x , and $k=3$ classes, you have a matrix of weights that is k by d .



note that w_1 has the same dimension as x .
if x is $28 \times 28 = 784$ pixels then
 w_1 is also a vector of length 784.

multi-class logistic regression, Cross entropy loss



Example:

Say $w_1 = [1, 0, 0]$, $w_2 = [0, 1, 0]$, $w_3 = [0, 1, 1]$, $x = [1, 2, 3]$, $y = \text{class 2}$.

Compute the scores, the probabilities of the logistic regression for these 3 classes and the loss.

scores = $[1, 2, 5]$, probabilities = $\text{softmax}(\text{scores})$

$= [\exp(1), \exp(2), \exp(5)] / C$

$C = \exp(1) + \exp(2) + \exp(5) = 158.5$

$= [2.7, 7.38, 148] / 158 =$

probabilities = $p = [0.01, 0.04, 0.94]$

loss = cross entropy of $[y, p]$

$= -\sum y_i \log p_i$

$y = [0, 1, 0]$ if $y = \text{class 2}$.

$= -\log(0.04) = 4.64$

If correct class was 3

$y = [0, 0, 1]$

loss = $-\log(0.94) = 0.09$

If logistic regression is sure and correct it pays loss:

$-\log 1 = 0$

if logistic regression is sure and wrong it pays loss:

$-\log 0 = +\text{infinity}$.

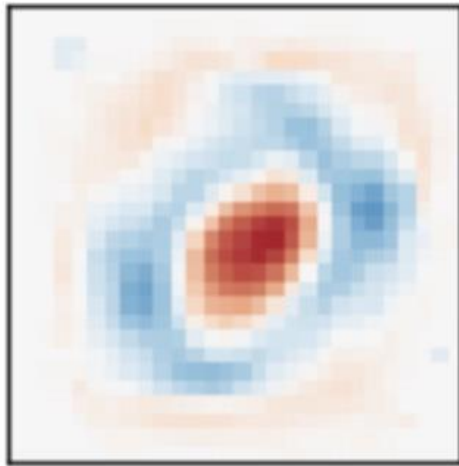
Observation on Cross Entropy loss

- $y = [1, 0, 0]$
- model1: $\text{softmax}(W_1 x) = [0.8 \ 0.1 \ 0.1]$
- model2: $\text{softmax}(W_2 x) = [0.8 \ 0 \ 0.2]$
- cross entropy loss for model1 **VS** cross entropy loss for model 2
- **Question:** Which model has higher loss?

Observation on Cross Entropy loss

- $y = [1, 0, 0]$
- model1: $\text{softmax}(W_1 x) = [0.8 \ 0.1 \ 0.1]$
- model2: $\text{softmax}(W_2 x) = [0.8 \ 0 \ 0.2]$
- the cross entropy for model1 = cross entropy loss for model 2
- the most straightforward way to give 'partial credit' is to 'soften' the true labels.
- $y_{\text{soft}} = [0.9 \ 0.1/(k-1) \ 0.1/(k-1)]$

visualizing w_0 :

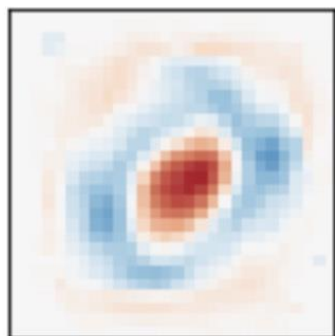


Class 0

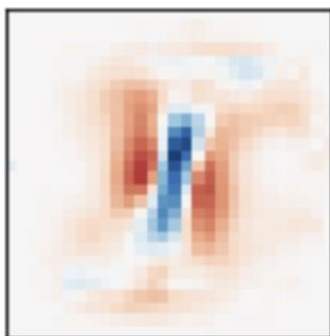
this is `model.coefs` reshaped as a 28 by 28 image

```
coefs = model.coef_.copy()  
plt.imshow(coefs[0].reshape(28,28) )  
plt.show()
```

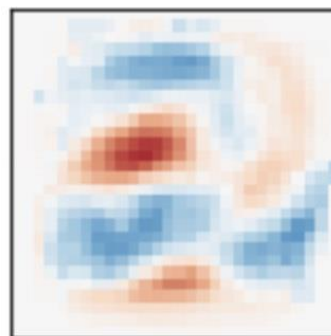
visualizing $w_0, w_1, w_2 \dots w_9$



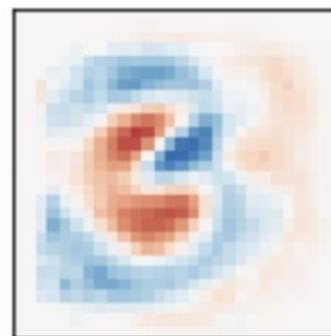
Class 0



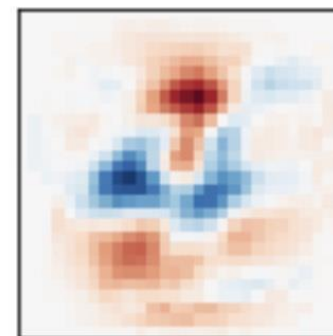
Class 1



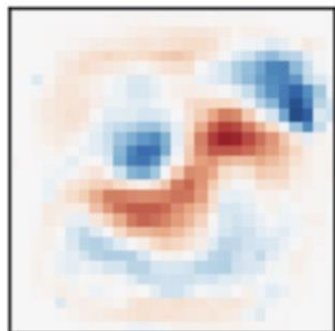
Class 2



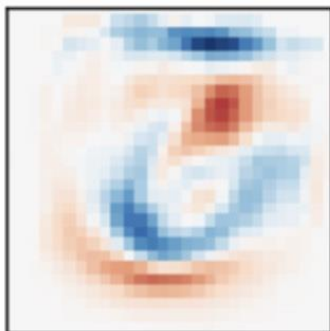
Class 3



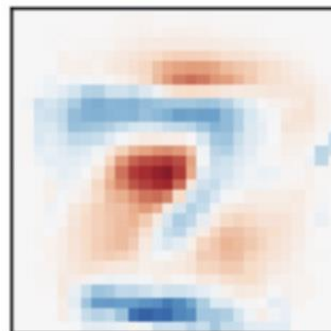
Class 4



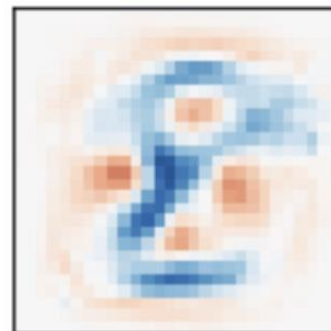
Class 5



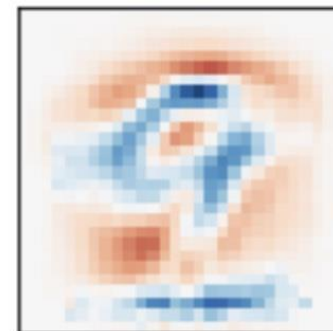
Class 6



Class 7



Class 8



Class 9

Fun Side:

How to compute the entropy of English

- There are 26 letters + space, so how many bits per letter?
- $\log 27 = 4.75$ bits per English letter.
- B*t th* En*l*sh l*ng***e i* c*mpre*sib*e
(44 characters, only 32 given)
- If we can make a generative model for English, we can use it for
 - Generating fake English text
 - Denoising
 - Compressing
 - Classification
 - Break cryptographic codes

How to generate English, zero-th order letters

- Pick randomly one letter from 27, uniformly
- X
- F
- O
- M
- L
- Space
- R
- XFOML RXKHRJFFJUJ ZLPWCFWKCYJ FFJEYVKCC

How to generate English, First order letters (match letter marginal distribution)

- Pick randomly one letter from 27, using marginal

- O

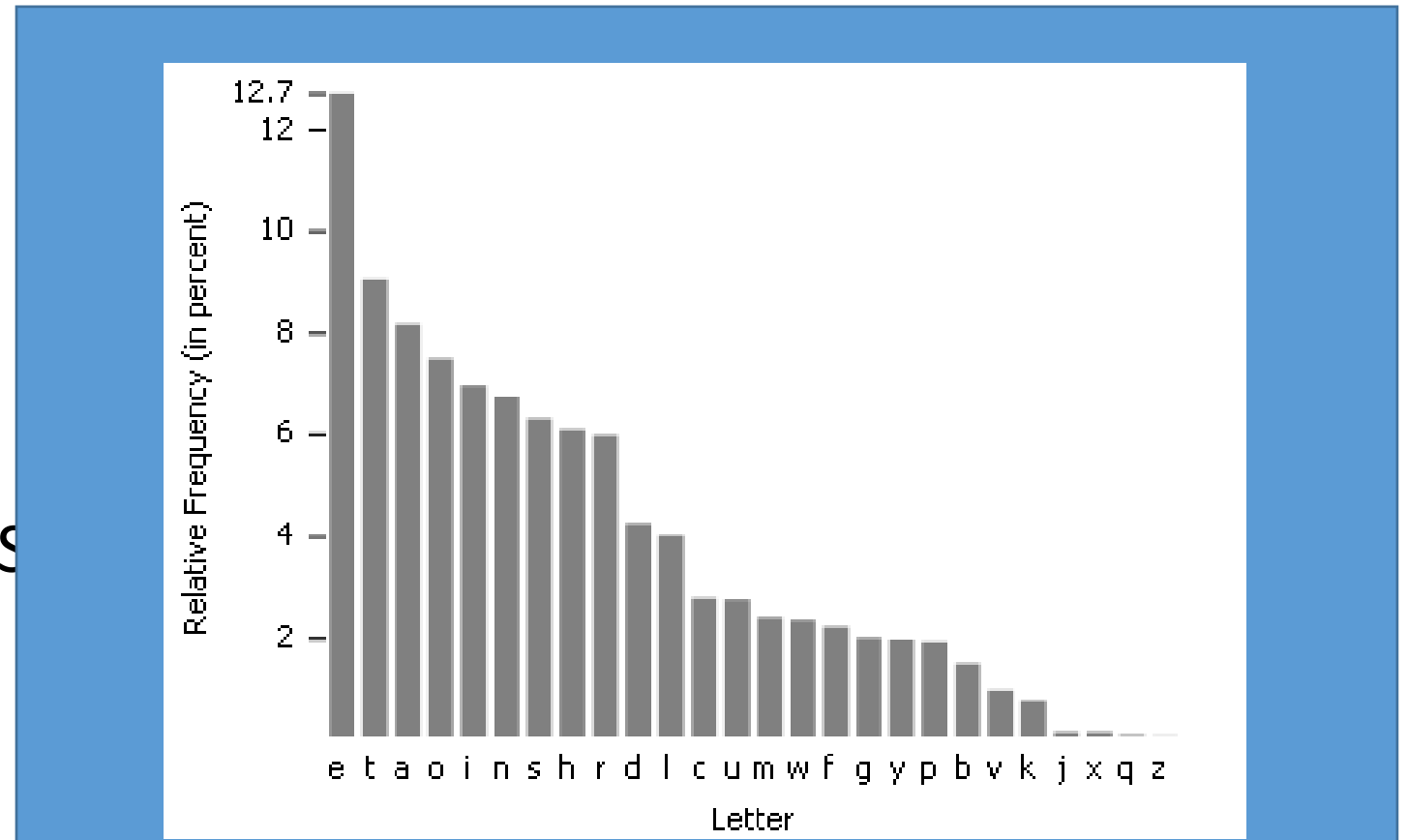
- C

- R

- O

- (space)

- OCRO HLI RGWR NMIELWIS
OOBTTVA NAH BRL



How to generate English, Second order Digrams (match letter conditional distribution)

- Pick randomly one letter from 27, using marginal
- O
- Pick second letter from prob (Letter 2 | letter 1 = O).
- Histogram of which letter follows O
- Exercise:
- Given this dataset, compute the marginals for O,N,B
Compute $\Pr(\text{letter } n / \text{letter } n-1 = \text{O})$.
- ON THE ONSET OF BOB

From Shannon's 1948 paper, 'A mathematical Theory of Communication'

1. Zero-order approximation (symbols independent and equiprobable).

XFOML RXKHRJFFJUJ ZLPWCFWKCYJ FFJEYVKCQSGHYD QPAAMKBZAACIBZL-HJQD.

2. First-order approximation (symbols independent but with frequencies of English text).

OCRO HLI RGWR NMIELWIS EU LL NBNESEBYA TH EEI ALHENHTTPA OOBTTVA NAH BRL.

3. Second-order approximation (digram structure as in English).

ON IE ANTSOUTINYS ARE T INCTORE ST BE S DEAMY ACHIN D ILONASIVE TU-COOWE AT TEASONARE FUSO TIZIN ANDY TOBE SEACE CTISBE.

4. Third-order approximation (trigram structure as in English).

IN NO IST LAT WHEY CRATICT FROURE BIRS GROCID PONDENOME OF DEMONSTURES OF THE REPTAGIN IS REGOACTIONA OF CRE.

5. First-order word approximation. Rather than continue with tetragram, $\dots$, n -gram structure it is easier and better to jump at this point to word units. Here words are chosen independently but with their appropriate frequencies.

REPRESENTING AND SPEEDILY IS AN GOOD APT OR COME CAN DIFFERENT NATURAL HERE HE THE A IN CAME THE TO OF TO EXPERT GRAY COME TO FURNISHES THE LINE MESSAGE HAD BE THESE.